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Diagonalization Graded Assignment
Graded Assignment • 10 min

Due Nov 11, 11:59 PM IST

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1. **True** or **False**: If an $n \times n$ matrix A has n distinct eigenvalues, then A is diagonalizable. 1 / 1 point

   - (•) True
   - ( ) False

   Nice work

2. **True** or **False**: If $A\mathbf{u}_1 = \lambda_1\mathbf{u}_1$ and $A\mathbf{u}_2 = \lambda_2\mathbf{u}_2$, where $\lambda_1 \neq \lambda_2$, and $\mathbf{u}_1$ and $\mathbf{u}_2$ are nonzero vectors, then $\{\mathbf{u}_1, \mathbf{u}_2\}$ is linearly independent. 1 / 1 point

   - (•) True
   - ( ) False

   Nice work

3. **True** or **False**: If A and B are 2×2 diagonalizable matrices, then AB is diagonalizable. 1 / 1 point

   - ( ) True
   - (•) False

   Nice work

4. **True** or **False**: If A is diagonalizable, then A^T is diagonalizable. 1 / 1 point

   - (•) True
   - ( ) False

   Nice work

5. **True** or **False**: The matrix $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ is diagonalizable. 1 / 1 point

   - ( ) True
   - (•) False

   Nice work